

Warehouse & Retail Sales Forecasting

Project Report — Data Preparation, Feature Engineering & Predictive Modeling
Dataset: Warehouse and Retail Sales.csv

1. Executive Summary

This report documents an end-to-end retail sales forecasting project built on monthly warehouse and retail transaction records. The workflow covers data cleaning, feature engineering (calendar, lag, and rolling-average features), and comparison of two forecasting models — Linear Regression and Random Forest Regression. Random Forest was identified as the stronger of the two models, though both models' accuracy was constrained by a very limited effective training window after filtering and missing-month gaps in the underlying data. Key data-quality limitations and recommended next steps are outlined in the closing sections.

Key Finding

The current model was trained on only 14 monthly observations (test set: 4 months) because the analysis window was restricted to dates before 2021-01-01, even though the raw dataset spans 2017–2026. Combined with 83 missing calendar months, this significantly limited model accuracy.

2. Dataset Overview

The raw dataset contains item-level warehouse and retail transaction records across multiple years.

Attribute	Value
Total records (raw)	341,037
Columns	YEAR, MONTH, SUPPLIER, ITEM CODE, ITEM DESCRIPTION, ITEM TYPE, RETAIL SALES, RETAIL TRANSFERS, WAREHOUSE SALES
Years covered	2017, 2018, 2019, 2020, 2026
Duplicate rows	0
Missing values (pre-clean)	SUPPLIER: 204 · ITEM TYPE: 1 · RETAIL SALES: 166 (after coercion)
Negative sales values	None detected in RETAIL SALES / RETAIL TRANSFERS / WAREHOUSE SALES

3. Data Preparation

The following cleaning steps were applied before modeling:

- Converted RETAIL SALES, RETAIL TRANSFERS, and WAREHOUSE SALES to numeric type, coercing invalid entries to NaN.
- Filled missing SUPPLIER and ITEM TYPE text fields with "Unknown".
- Filled missing numeric sales values with 0.
- Constructed a proper DATE column (YEAR-MONTH-01) and converted it to datetime64 format.
- Sorted all records chronologically by DATE.
- Applied an IQR-based outlier scan on the three sales columns (see below); outliers were flagged but retained in the dataset.
- Aggregated item-level records into a monthly time series by summing RETAIL SALES per DATE.

IQR Outlier Scan Results

Column	Outliers Detected
RETAIL SALES	53,127
RETAIL TRANSFERS	57,066
WAREHOUSE SALES	47,890

Data Coverage Issue

After aggregating to a monthly series, a calendar-completeness check revealed **83 missing months** between June 2017 and August 2026 — including nearly all of 2021 through 2026. In addition, the modeling step further restricted the dataset to records before 2021-01-01, leaving only 24 monthly observations for feature engineering. This combination is the primary driver of the limited model performance reported in Section 6.

4. Feature Engineering

The following predictive features were engineered from the monthly retail sales series:

Feature	Description
MONTH / QUARTER / YEAR	Calendar components extracted from DATE
DAY_OF_WEEK	Day-of-week of the 1st of each month
SEASON	Winter / Spring / Summer / Autumn indicator derived from MONTH
LAG_1, LAG_2, LAG_3	Retail sales from 1, 2, and 3 months prior
ROLLING_MEAN_3	Trailing 3-month average of retail sales
ROLLING_MEAN_6	Trailing 6-month average of retail sales

No promotional or discount variables were engineered, since the source data does not contain campaign information — synthetic promotional flags were deliberately avoided to prevent introducing fabricated signal.

Rows with insufficient history for lag/rolling calculations were dropped, reducing the feature-engineered dataset to **18 monthly observations**. The data was then split chronologically (not randomly, to preserve time-series ordering) into training and test sets.

Train / Test Split

Split	Rows	Period
Training	14	2017-12-01 to 2019-11-01
Testing	4	2020-01-01 to 2020-09-01

5. Model Development

Two regression models were trained on the engineered feature set (MONTH, QUARTER, YEAR, DAY_OF_WEEK, SEASON, LAG_1-3, ROLLING_MEAN_3, ROLLING_MEAN_6) to predict monthly RETAIL_SALES:

- **Linear Regression** — a baseline linear model capturing straight-line relationships between features and sales.
- **Random Forest Regression** (100 trees, random_state=42) — an ensemble model capable of capturing non-linear patterns.

6. Model Evaluation

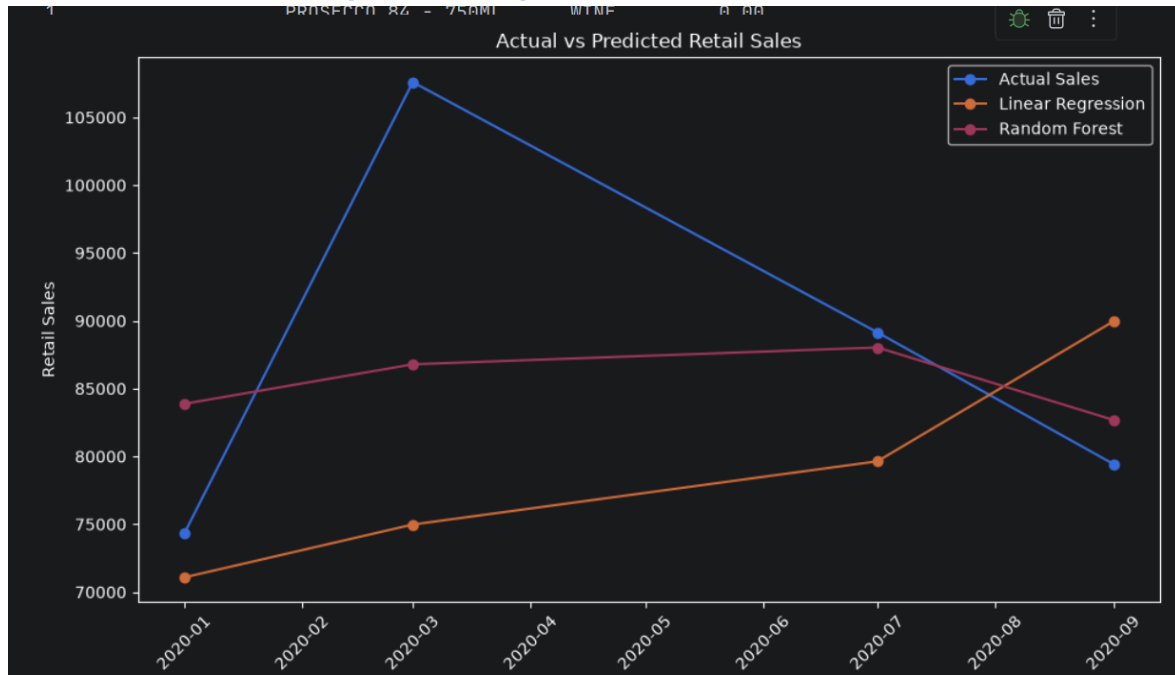
Model accuracy was assessed on the held-out test period (Jan–Sep 2020) using MAE, RMSE, and R² score.

Model	MAE	RMSE	R ² Score
Linear Regression	13,973.30	17,858.01	-0.977
Random Forest	8,681.32	11,575.74	0.169

Random Forest was the best-performing model on all three metrics (lowest MAE, lowest RMSE, highest R²). However, a negative R² for Linear Regression indicates it performed worse than simply predicting the average sales value — a strong signal that the training set was too small and too discontinuous (see Section

3) for either model to learn a robust pattern. Random Forest's R^2 of 0.169, while an improvement, still reflects weak explanatory power for production use.

Actual vs. Predicted Sales (Test Period)



Actual sales (blue) show a sharp spike in March 2020 that neither model anticipated. Linear Regression (orange) tracks a near-monotonic upward trend disconnected from actual movement, while Random Forest (pink) stays in a narrow band, effectively predicting close to the historical average rather than capturing volatility.

7. Seasonal Patterns

Average retail sales by calendar month (computed on the historical training window) show a clear year-end peak and a post-holiday trough:

Month	Avg. Retail Sales
December	128,848.57
November	98,346.96
March	96,018.13
May	92,508.26
June	90,779.80
July	88,884.51
August	87,286.89
October	86,874.24
September	82,933.65
April	79,051.38
February	78,360.73
January	75,404.65

Highest-selling month: December (holiday-season peak) | **Lowest-selling month:** January (typical post-holiday slowdown)

8. Business Insights

Important Factors Affecting Sales

- Lagged sales and rolling averages carry predictive signal, since recent sales activity tends to influence near-term future sales.
- Seasonal indicators (month, season) help capture recurring, cyclical demand patterns such as the December peak.

Inventory & Marketing Implications

- Build up inventory ahead of historically strong periods (November–December) to avoid stockouts.
- Use lower-demand periods (January–February) to trim excess inventory and reduce carrying costs.
- Plan marketing campaigns to land just before seasonal peaks; use promotions to stimulate demand in slow months.
- Feed forecast output into procurement and resource-allocation planning, refreshing the model as new data arrives.

Historical Sales Summary

Metric	Month	Value
Highest monthly sales	December 2017	128,848.57
Lowest monthly sales	January 2020	74,319.53

9. Limitations

- **Restricted date window:** the modeling step filtered data to before 2021-01-01, discarding all 2021–2026 records and leaving only 18 usable rows after feature engineering.
- **Missing months:** 83 calendar months are absent from the aggregated series, which distorts lag features (a "1-month-prior" value may actually reflect a gap of several months).
- **Small test set:** only 4 test-period observations were available, making evaluation metrics statistically fragile.
- **Outliers retained:** over 47,000–57,000 line-item outliers were flagged but not removed or capped prior to monthly aggregation.

10. Recommendations & Next Steps

Recommended Actions

1. Remove the 2021-01-01 date filter and use the full 2017–2026 history to train on a much larger sample.
2. Reindex the monthly series to fill calendar gaps before computing lag/rolling features, so they reflect true month-over-month relationships.
3. Re-run outlier treatment (cap or investigate extreme line items) prior to aggregation.
4. Re-evaluate Linear Regression and Random Forest on the corrected dataset, and consider adding a dedicated time-series model (e.g., SARIMA or Prophet) for comparison.
5. Establish a recurring refresh cycle so forecasts incorporate new sales data as it becomes available.